RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Additional Practice Questions for Chapter S6: Correlation and Regression

1 Explain why it is advisable to draw a scatter diagram before interpreting a correlation coefficient calculated for a sample drawn from a bivariate distribution. [1]

Sketch a scatter diagram that shows an obvious relation between two variables but which yields a linear product-moment correlation coefficient that is close to zero. [1]

2 Seven cities in a certain country are linked by rail to the capital city. The table below shows the distance of each city from the capital and the rail fare from the city to the capital.

City	*A*	*B*	*C*	*D*	*E*	*F*	*G*
Distance, x km	124	44	76	148	16	180	104
Rail fare, \$ y	156	53	99	169	23	177	138

(i) Give a sketch of the scatter diagram for the data as shown on your calculator. [2]

(ii) Calculate the product moment correlation coefficient. [1]

You are given that the regression line of y on x has equation $y = 16.7 + 1.01x$, where the coefficients are given correct to 3 significant figures.

(iii) Calculate the equation of the regression line of x on y, giving your answer in the form $x = a + by$. [1]

(iv) Use the appropriate regression line to estimate

(a) the rail fare from a city that is 28 km from the capital, [2]

(b) the distance of a city from the capital if the rail fare is \$198. [2]

(v) Comment briefly on the reliability of the estimates in part **(iv)**. [2]

[**(ii)** $r = 0.973$ (3 s.f.) **(iii)** $x = -10.6 + 0.940y$ (3 s.f.) **(iv)(a)** \$45 **(b)** 176 km]

3 An engineering company makes cranes. The numbers, x, sold in each three-month period for two years, together with the profits, y thousand dollars, on the sale of these cranes are given in the following table.

x	15	17	13	21	16	22	14	18
y	290	350	270	430	340	410	300	360

(i) Give a sketch of the scatter diagram for the data as shown in your calculator. [2]

(ii) Find $\bar{x}$ and $\bar{y}$, and mark the point $(\bar{x}, \bar{y})$ on your scatter diagram. [2]

(iii) Calculate the equation of the regression line of y on x, and draw this line on your scatter diagram. [2]

(iv) Calculate the product moment correlation, and comment on its value in relation to your scatter diagram. [2]

(v) For the next three-month period, the sales target is 20 cranes. Estimate the corresponding profit. [2]

(vi) The company's sales director uses the regression line in **(iii)** to predict the profit if 40 cranes were to be sold in a three-month period. Comment on the validity of this prediction. [2]

[**(ii)** $\bar{x} = 17, \bar{y} = 344$ **(iii)** $y = 17.1x + 53.3$ **(iv)** 0.969 **(v)** \$395,000]

4 A medical officer wishes to investigate a patient's heart-beat rate h beats per minute (bpm) when he walks at different speeds s km/h. The data is shown below:

s (km/h)	1	1.5	2	2.5	3	3.5	4	4.5	5
h (bpm)	60	63	66	75	86	99	150	110	130

(i) Sketch a scatter plot of the above data. [1]

(ii) One of the values of h appears to be incorrect. Indicate the corresponding point on your diagram by labelling it P. [1]

Omit P for the remainder of this question.

(iii) Calculate the product moment correlation coefficient for this set of data. Use the equation of an appropriate regression line to predict the value of s when $h = 100$, justifying your choice of regression line. [4]

It is suggested to use one of the following two models instead:

Model (I): $h = a + bs^2$,

Model (II): $h = a + b\mathrm{e}^s$,

where a and b are real constants.

(iv) Determine which of the two models is a better choice, giving a reason for your answer. [2]

(v) Suppose a new data pair $(\bar{s}, \bar{h})$ is added to the table above, where $\bar{s}$ and $\bar{h}$ are the patient's sample mean walking speed (in km/h) and his sample mean heart-beat rate (in bpm) respectively, based on the data above. Without any calculations, explain whether the equation of the regression line you have obtained in part **(iii)** would change. [2]

[**(iii)** 0.981, $s = 3.67$ **(iv)** model (I)]

5 **(a)** Sketch a scatter diagram that might be expected when x and y are related approximately by $y = px^2 + t$ in each of the cases **(i)** and **(ii)** below. In each case your diagram should include 6 points, approximately equally spaced with respect to x, and with all x-values positive.

(i) p and t are both positive.

(ii) p is negative and t is positive. [2]

(b) The ages in months (m) and prices in dollars (P) of a random sample of ten used cars of a certain model are given in the table.

m	11	20	28	36	40	47	58	62	68	75
P	112800	102600	76500	72000	72000	69000	65800	57000	50600	47600

It is thought that the price after m months can be modelled by one of the formulae

$$P = am + b, \qquad P = c\ln m + d,$$

where a, b, c and d are constants.

(i) Find, correct to 4 decimal places, the value of the product moment correlation coefficient between

(A) m and P,

(B) $\ln m$ and P. [2]

(ii) Explain which of $P = am + b$ and $P = c\ln m + d$ is the better model and find the equation of a suitable regression line for this model. [3]

(iii) Use the equation of your regression line to estimate the price of a car that is 50 months old. [1]

[**(b)(i)** -0.9470, -0.9749 **(ii)** $P = 196000 - 33700\ln m$ **(iii)** \$64000]

6 Research is being carried out into how the concentration of a drug in the bloodstream varies with time, measured from when the drug is given. Observations of successive times give the data shown in the following table.

Time (t minutes)	15	30	60	90	120	150	180	240	300
Concentration (x micrograms per litre)	82	65	43	37	22	19	12	6	2

It is given that the value of the product moment correlation coefficient for this data is -0.912, correct to 3 decimal places. Draw a scatter diagram for the data. [1]

Calculate the equation of the regression line of x on t. [2]

Calculate the corresponding estimated value of x when $t = 300$, and comment on the suitability of the linear model. [2]

The variable y is defined by $y = \ln x$. For the variables y and t,

(i) calculate the product moment correlation coefficient and comment on its value, [2]

(ii) calculate the equation of the appropriate regression line. [3]

Use a regression line to give the best estimate that you can of the time when the drug concentration is 15 micrograms per litre. [2]

[$x = 66.2 - 0.260t$, -11.7 **(i)** $r = -0.994$ **(ii)** $y = 4.62 - 0.0123t$, 155]

7 Mr. Lee recorded the length of time, y minutes, taken to travel to work when leaving home x minutes after 6.00 am on seven selected mornings. The results are as follows:

x /min	0	10	20	30	40	50	60
y /min	16	27	28	39	43	48	51

(i) Calculate the equations of the least squares regression lines of y on x and x on y. [2]

(ii) Mr. Lee forgot to add an observation for a particular morning but remembered that the two regression lines remain unchanged on its inclusion. Find this missing observation. [1]

Mr. Lee needs to report to work by 8.00 am. He leaves home x minutes after 6.00 am and he arrives at his workplace z minutes before 8.00 am.

(iii) Find z in terms of x. [1]

(iv) Hence estimate, to the nearest minutes, the latest time Mr. Lee can leave home without arriving late at work. [2]

[**(i)** $y = 0.579x + 18.6$, $x = 1.67y - 30$ **(ii)** (30,36) **(iii)** $z = 101 - 1.58x$ **(iv)** 7.04 am]

8 A website about electric motors gives information about the percentage efficiency of motors depending on their power, measured in horsepower. Xian has copied the following table for a particular type of electric motor, but he has copied one of the efficiency values wrongly.

Power, x	1	1.5	2	3	5	7.5	10	20	30	40	50
Efficiency, y%	72.5	82.5	84.0	87.4	87.5	88.5	89.5	90.2	91.0	91.7	92.4

(i) Plot a scatter diagram on graph paper for these values, labeling the axes, using a scale of 2 cm to represent 10% efficiency on the y-axis and an appropriate scale for the x-axis. On your diagram, circle the point that Xian has copied wrongly. [2]

For parts **(ii)**, **(iii)** and **(iv)** of this question you should **exclude** the point for which Xian has copied the efficiency value wrongly.

(ii) Explain from your scatter diagram why the relationship between x and y should not be modelled by an equation of the form $y = ax + b$. [1]

(iii) Suppose that the relationship between x and y are modelled by an equation of the form $y = \frac{c}{x} + d$, where c and d are constants. State with a reason whether each of c and d is positive or negative. [2]

(iv) Find the product moment correlation coefficient and the constants c and d for the model in part **(iii)**. [3]

(v) Use the model $y = \frac{c}{x} + d$ with the values of c and d found in part **(iv)**, to estimate the efficiency value (y) that Xian has copied wrongly. Give two reasons why you would expect this estimate to be reliable. [3]

[**(iv)** -0.980, $c = -17.5$ and $d = 91.8$ **(v)** 85.9]

9 For 3 years a factory records the number of units it produces each quarter and the total cost of producing the units. The following table summarises these results.

Units of output (x) in 1000's	15	30	57	75	10	25	49	70	20	34	60	76
Total cost (y) in £1000's	33	49	71	99	34	40	61	82	40	51	84	93

(i) Give a sketch of the scatter diagram for the data as displayed on your graphing calculator. [2]

(ii) Calculate the equation of the regression line of y on x, draw this line on your diagram and calculate the product moment correlation coefficient. [3]

The selling price of each unit of output is £1.80.

(iii) Use your graph to estimate the level of output (correct to 2 significant figures) at which the total cost and the total income balance. [2]

(iv) Give a brief interpretation of what this value means. [1]

[**(ii)** $y = 19.3 + 0.970x$, $r = 0.983$, **(iii)** 23000 units]

10 In an experiment the following information was gathered about air pressure P, measured in inches of mercury, at different heights above sea-level h, measured in feet.

h	2000	5000	10000	15000	20000	25000	30000	35000	40000	45000
P	27.8	24.9	20.6	16.9	13.8	11.1	8.89	7.04	5.52	4.28

(i) Draw a scatter diagram for these values, labelling the axes. [1]

(ii) Find, correct to 4 decimal places, the product moment correlation coefficient between

(a) h and P,

(b) $\ln h$ and P,

(c) $\sqrt{h}$ and P. [3]

(iii) Using the most appropriate case from part **(ii)**, find the equation which best models air pressure at different heights. [3]

(iv) Given that 1 metre = 3.28 feet, re-write your equation from part **(iii)** so that it can be used to estimate the air pressure when the height is given in metres. [2]

[**(ii)** -0.9807, -0.9748, -0.9986; **(iii)** $P = 34.8 - 0.147\sqrt{h}$; **(iv)** $P = 34.8 - 0.266\sqrt{h}$]

11 **(i)** Sketch a scatter diagram that might be expected when x and y are related approximately as given in each of the cases (A), (B) and (C) below. In each case your diagram should include 6 points approximately equally spaced with respect to x, and with all x- and y-values positive. The letters a, b, c, d, e and f represent constants.

(A) $y = a + bx^2$, where a is positive and b is negative,

(B) $y = c + d \ln x$, where c is positive and d is negative,

(C) $y = e + \dfrac{f}{x}$, where e is positive and f is negative. [3]

A motoring website gives the following about the distance travelled, y km, by a certain type of car at different speeds, x kmh^{-1}, on a fixed amount of fuel.

Speed, x	88	96	104	112	120	128
Distance, y	148	147	144	138	126	107

(ii) Draw the scatter diagram for these values, labelling the axes. [1]

(iii) Explain which of the three cases in part **(i)** is the most appropriate for modelling these values, and calculate the product moment correlation coefficient for this case. [2]

(iv) It is required to estimate the distance travelled at a speed of 110 kmh^{-1}. Use the case that you identified in part **(iii)** to find the equation of a suitable regression line, and use your equation to find the required estimate. [3]

[**(iii)** model (A), -0.939]

12 A car is placed in a wind tunnel and the drag force F for different wind speeds v, in appropriate units is recorded. The results are shown in the table.

v	0	4	8	12	16	20	24	28	32	36
F	0	2.5	5.1	8.8	11.2	13.6	17.6	22.0	27.8	33.9

(i) Draw the scatter diagram for these values, labelling the axes clearly. [2]

It is thought that the drag force F can be modelled by one of the formulae

$$F = a + bv \quad \text{or} \quad F = c + dv^2$$

where a, b, c and d are constants.

(ii) Find, correct to 4 decimal places, the value of the product moment correlation coefficient between

(a) v and F,

(b) v^2 and F. [2]

(iii) Use your answers to parts **(i)** and **(ii)** to explain which of $F = a + bv$ or $F = c + dv^2$ is the better model. [1]

(iv) It is required to estimate the value of v for which $F = 26.0$. Find the equation of a suitable regression line, and use it to find the required estimate. Explain why neither the regression line of v on F nor the regression line of v^2 on F should be used. [4]

[**(ii)** 0.9860, 0.9907]

13 In a study to determine the relationship between the temperature T (measured in degrees Celsius) and the altitude H (height above sea level, measured in metres) for a certain mountain, a team of meteorologists ascended and descended the mountain on a day with clear weather and obtained readings of T and H.

The table below shows the 10 readings obtained:

H	2300	2390	2626	2631	2700	3126	3131	3400	3450	3580
T	11.80	9.26	10.29	7.99	9.52	5.20	7.26	3.72	4.43	2.31

(a) On the same axes, sketch a scatter diagram of the data and draw the regression line of T against H. [2]

(b) State the product moment correlation coefficient between T and H. Comment, in context, on the value obtained. [2]

It was found that 5 of these 10 readings were recorded between 2 am and 5.30 am when the team of meteorologists were ascending the mountain, while the other 5 were recorded between 8 am and 11.30 am when they were descending. The routes used for ascending and descending were the same. Between 5.30 am and 8 am, the meteorologists remained on the summit.

For these readings, the regression equation for ascending, corrected to 3 significant figures, is $T = 23.1 - 0.00574H$. The regression equation for descending is $T = a + bH$.

(c) Find, correct to 3 significant figures, the values of a and b. [2]

One of the meteorologists recorded the temperature near the foot of the mountain as 26 degrees Celsius.

(d) Determine whether it is suitable to use any of the regression equations above to estimate the altitude of the foot of the mountain. [1]

A temperature of F degrees Fahrenheit is equivalent to a temperature of T degrees Celsius, where $F = \frac{9}{5}T + 32$.

(e) The regression equation for ascending, $T = 23.1 - 0.00574H$ can be written in the form $F = c + dH$, where c and d are constants. Find the values of c and d. [2]

(f) Use the equation found in part **(e)** to estimate the temperature in degrees Fahrenheit when the meteorologists were at an altitude of 2800 m while ascending. Give two reasons why you would expect this estimate to be reliable. [3]

[**(b)** $r = -0.948$ **(c)** $a = 26.7$, $b = -0.00633$ **(e)** $c = 73.58$, $d = -0.010332$ **(f)** $F = 44.6504$]

THE END